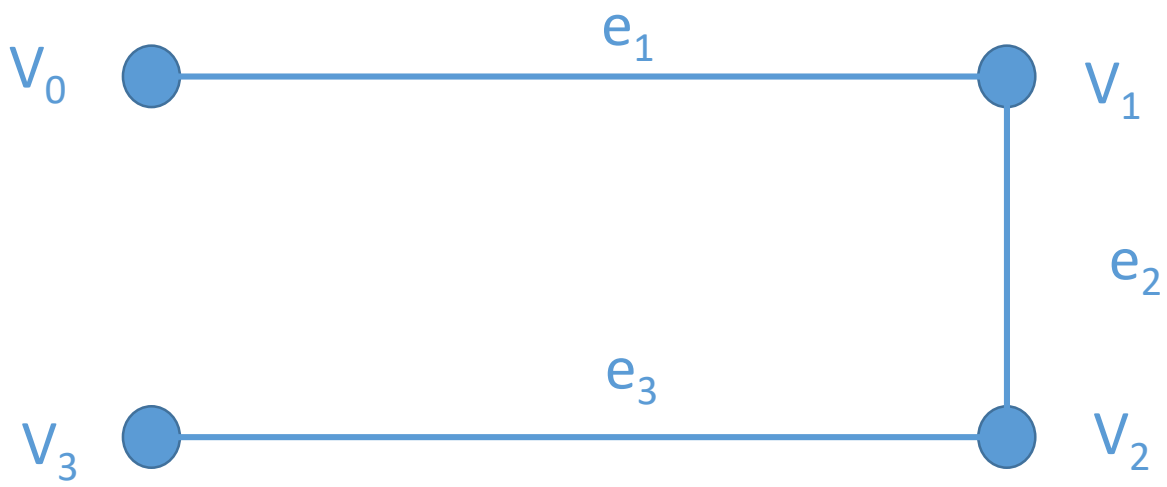
Path and Connectivity

Paths (Undirected)

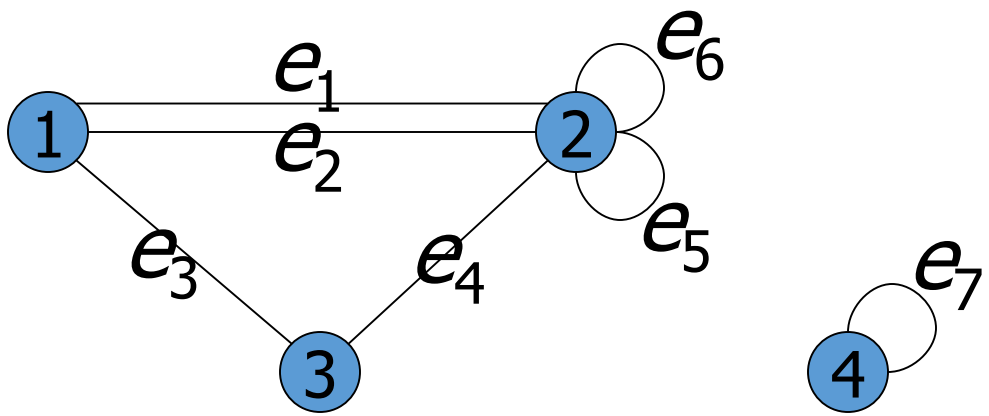
Definition

A ***path*** of length n in an undirected graph is a sequence of n edges $e_1, e_2, .., e_n$ such that each consecutive pair e_i , e_{i+1} share a (at least) common vertex



Paths (Undirected)

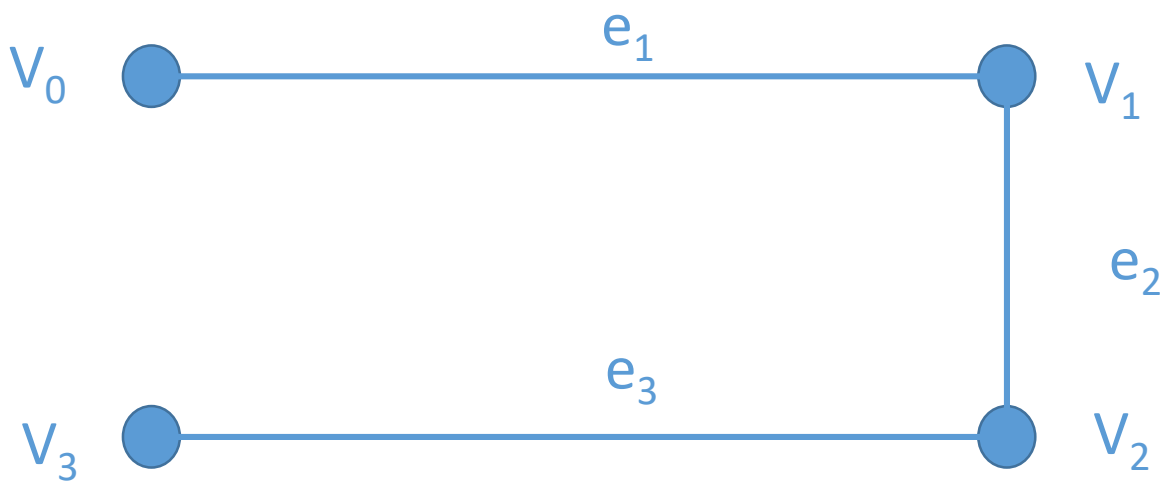
A path in a graph is a continuous way of getting from one vertex to another by using a sequence of edges.



Paths (Undirected)

Definition

Here the path from v_0 to v_3 is said to ***pass through*** the vertices v_0, v_1, v_2, v_3 or ***traverse*** the edges e_1, e_2, e_3.



Paths

Simple Path

A ***path or circuit is simple*** if it does not contain the same edge more than once. (though duplicate vertices are allowed).

A cycle (or circuit)

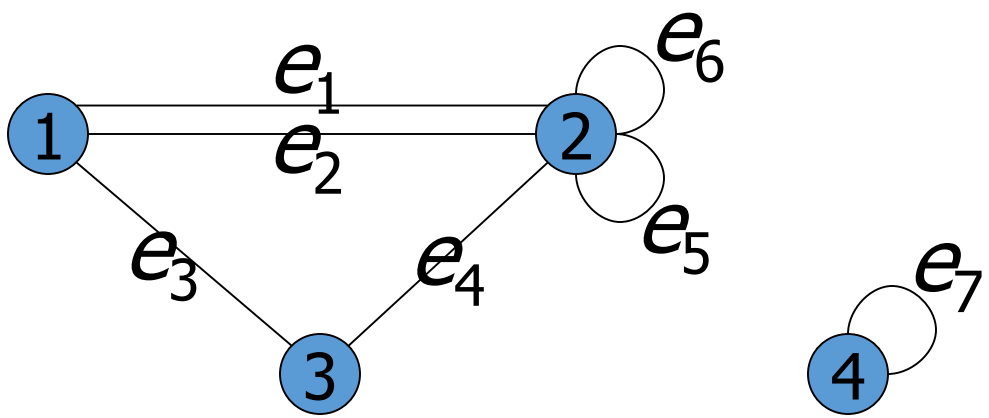
A ***cycle*** (or ***circuit***) is a path which starts and ends at the same vertex.

Isolated

A *path of 0 length has 1 vertex with 0 edge.*

Paths

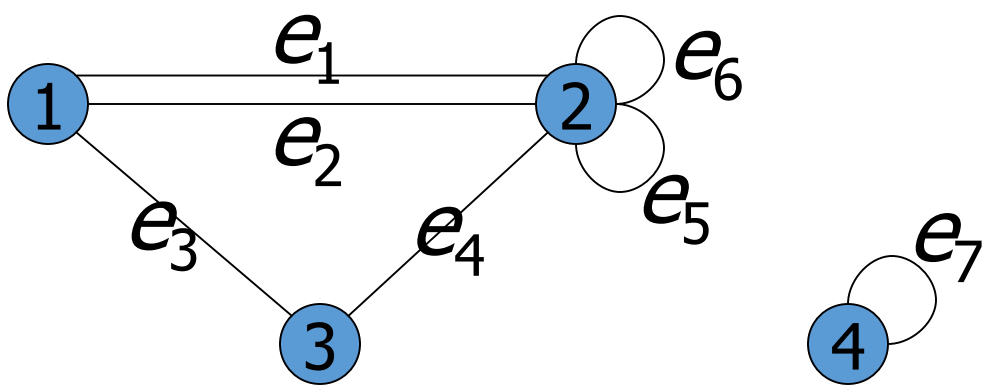
Q: Find a longest possible simple path in the following graph:



Paths

A: The following path from 1 to 2 is a maximal simple path because

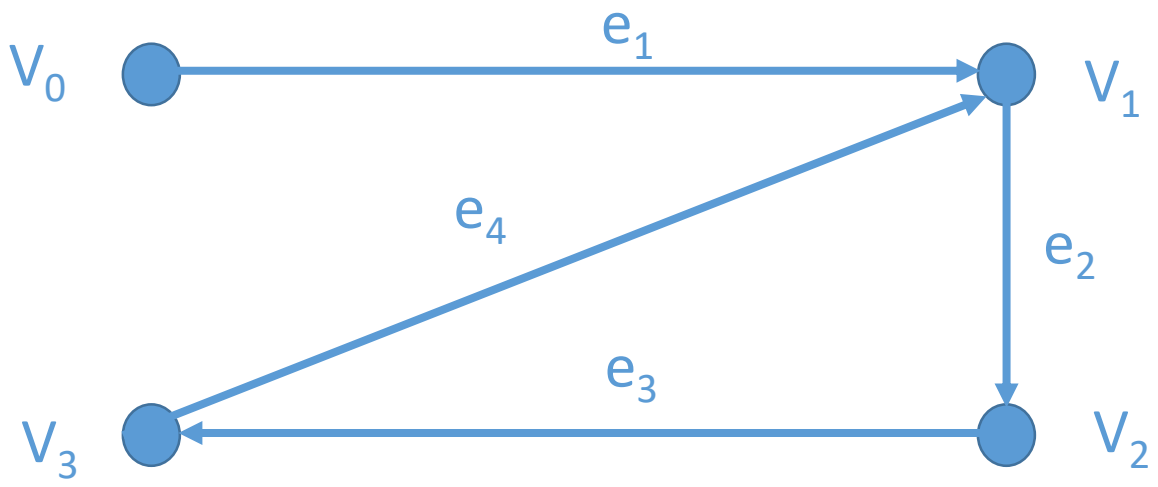
- simple: each of its edges appears exactly once
- maximal: because it contains every edge except the unreachable edge e_7



The maximal path: e_1, e_5, e_6, e_2, e_3, e_4

Paths (Directed)

A ***path*** of length n in a directed graph is a sequence of n edges $e_1, e_2, \ldots, e_n$ such that the target of e_i is the source e_{i+1} for each i.

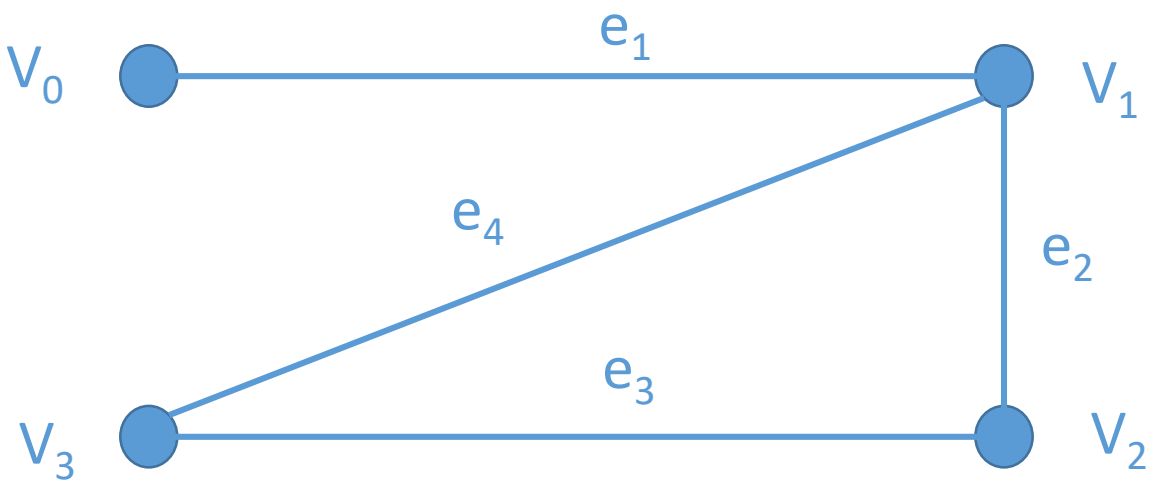


Connectedness in Undirected Graphs

Definition

An undirected graph is called connected if there is a path between every pair of distinct vertices of the graph. An undirected graph that is not connected is called disconnected.

Is it Connected?



Connected Component

Connectedness in Undirected Graphs

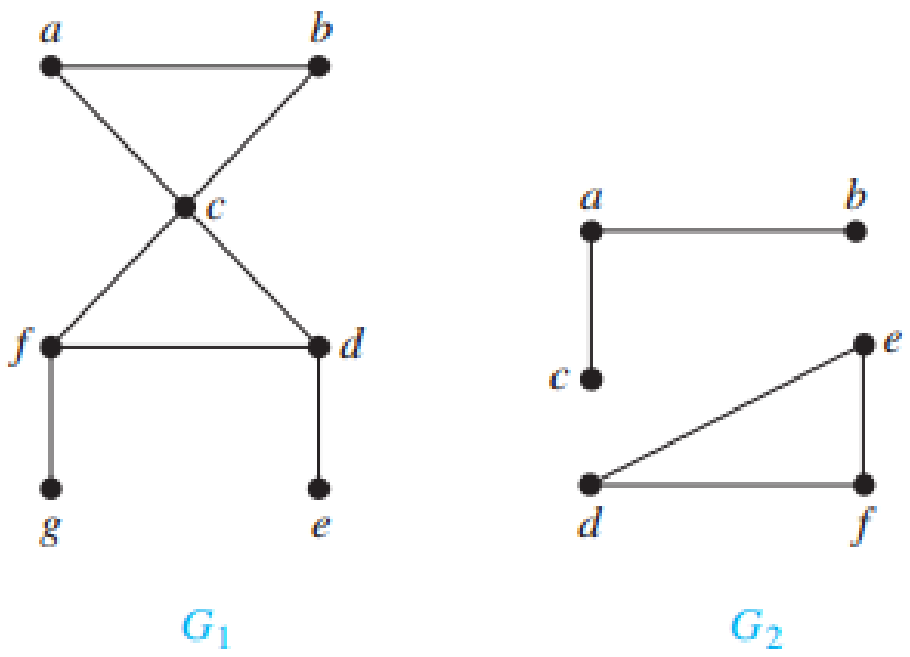
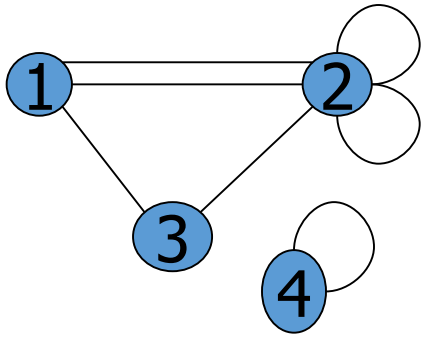
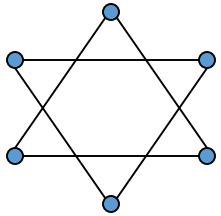
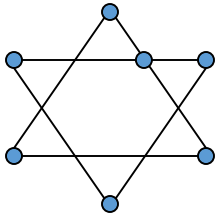


FIGURE 2 **The Graphs G_1 and G_2.**

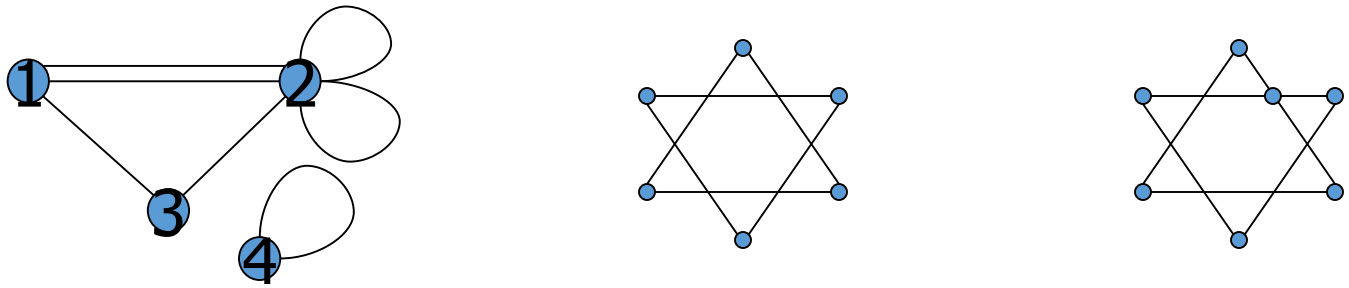
Q: Which of the following graphs are connected?







A: First and second are disconnected. Last is connected.



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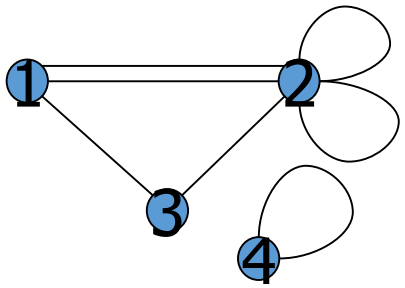

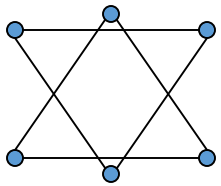

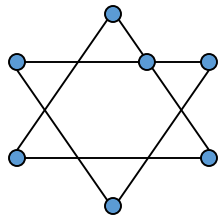

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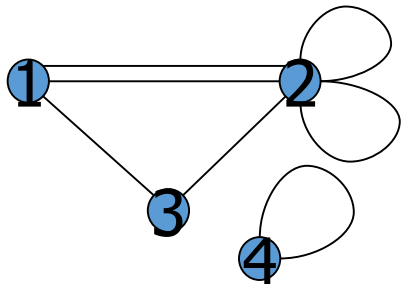

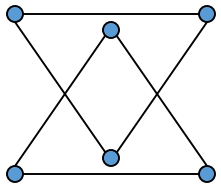

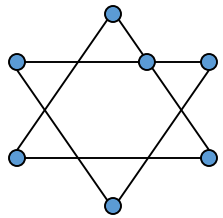

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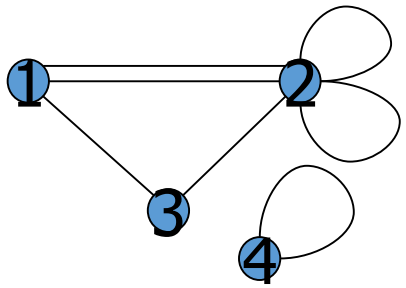

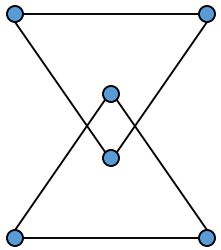

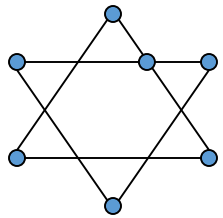

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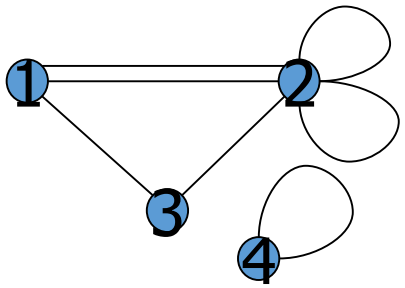

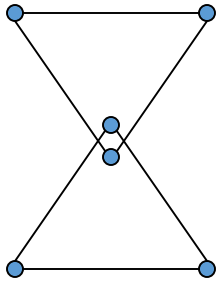

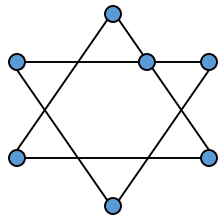

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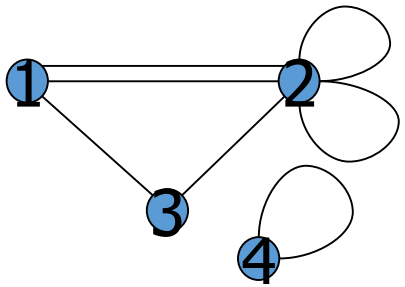
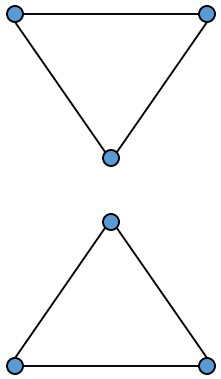
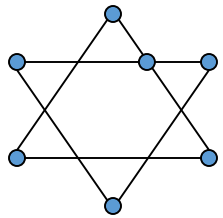




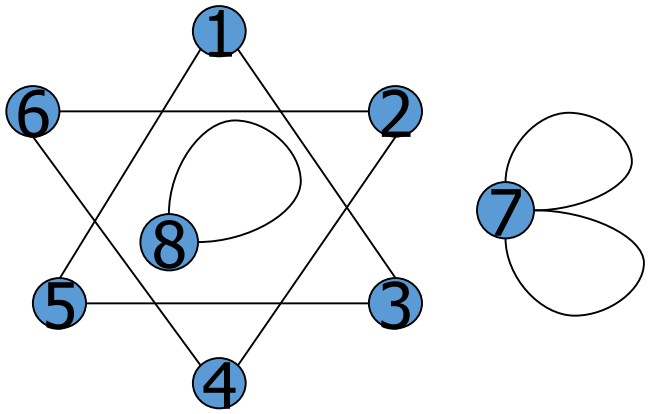


Connected Component

Definition

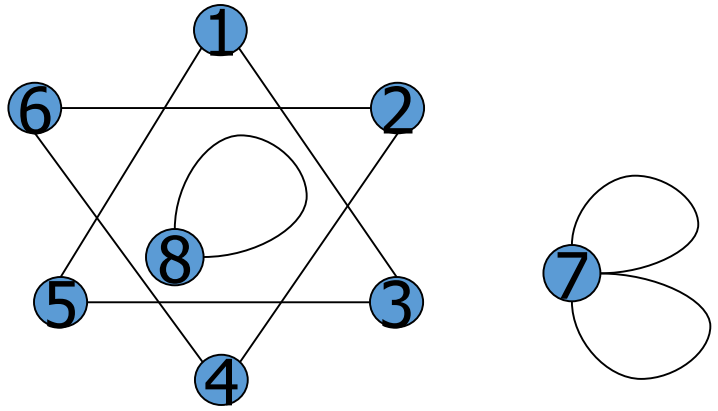
A ***connected component*** (or just ***component***) in a graph *G* is a set of vertices such that all vertices in the set are connected to each other and every possible connected vertex is included.

Q: What are the connected components of the following graph?



Connected Component

A: The components are {1,3,5},{2,4,6},{7} and {8} as one can see visually by pulling components apart:



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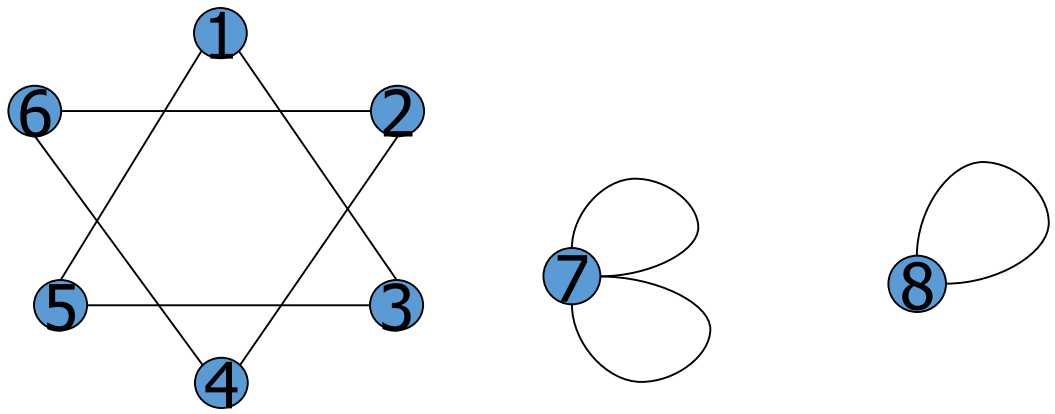

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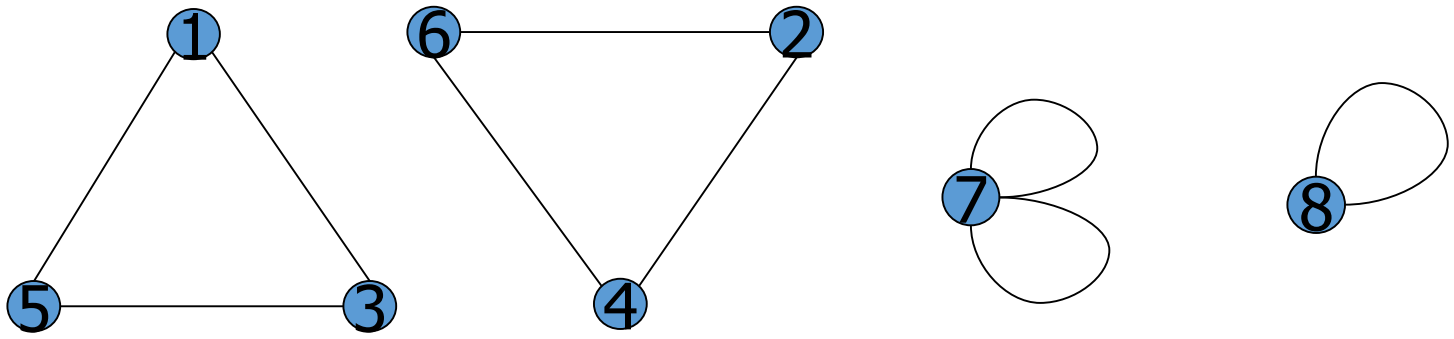


Euler Path and Circuit

Old City Old Problem!

The town of Königsberg, Prussia (now called Kaliningrad and part of the Russian republic), was divided into four sections by the branches of the Pregel River. In the eighteenth century seven bridges connected these regions.

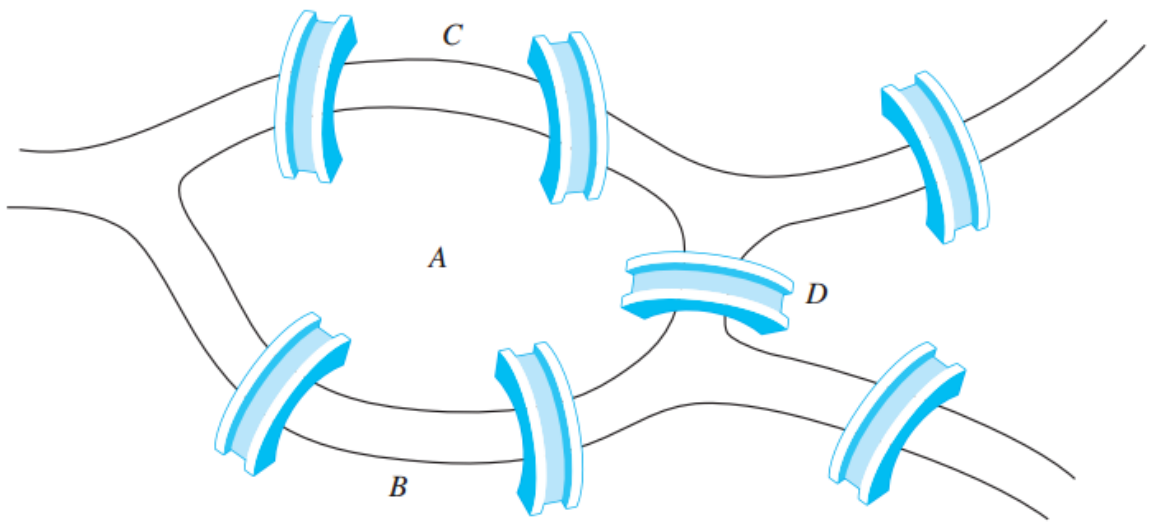
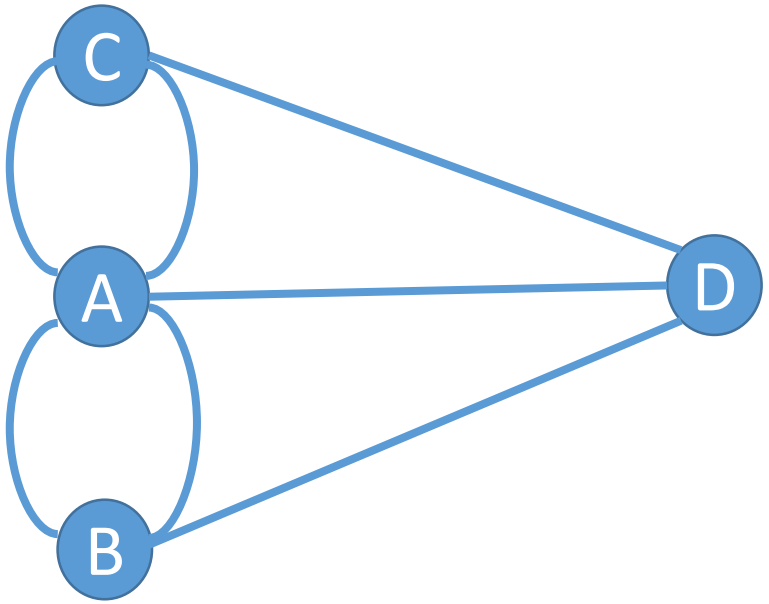


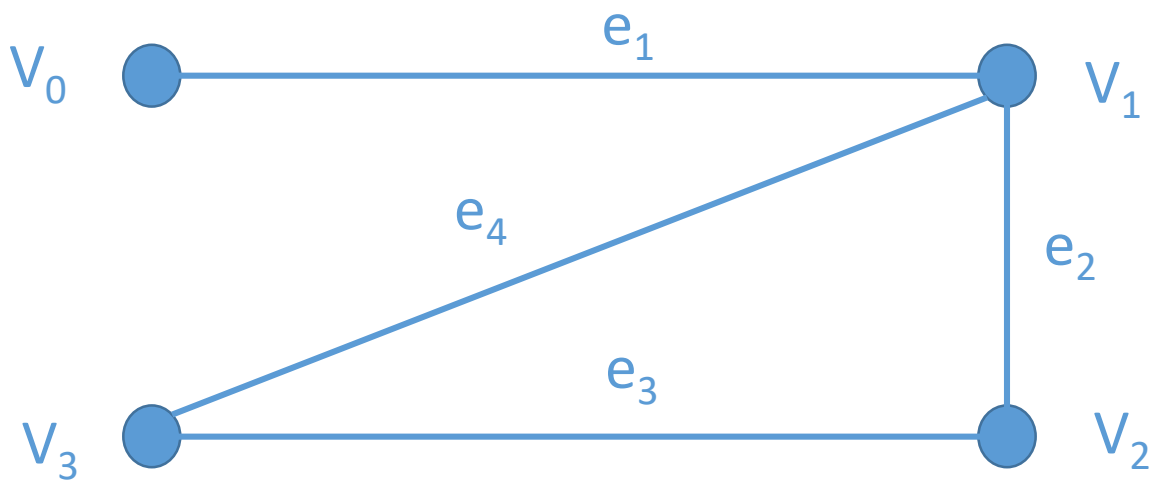
FIGURE 1 **The Seven Bridges of Königsberg.**



Euler circuit and Euler Path

Definition

- An ***Euler circuit*** an Eulerian circuit traverses every edge exactly once, but may repeat vertices.



Euler circuit and Euler Path

Theorem 8.3 (Euler):

A finite connected graph is Eulerian if and only if each vertex has even degree.

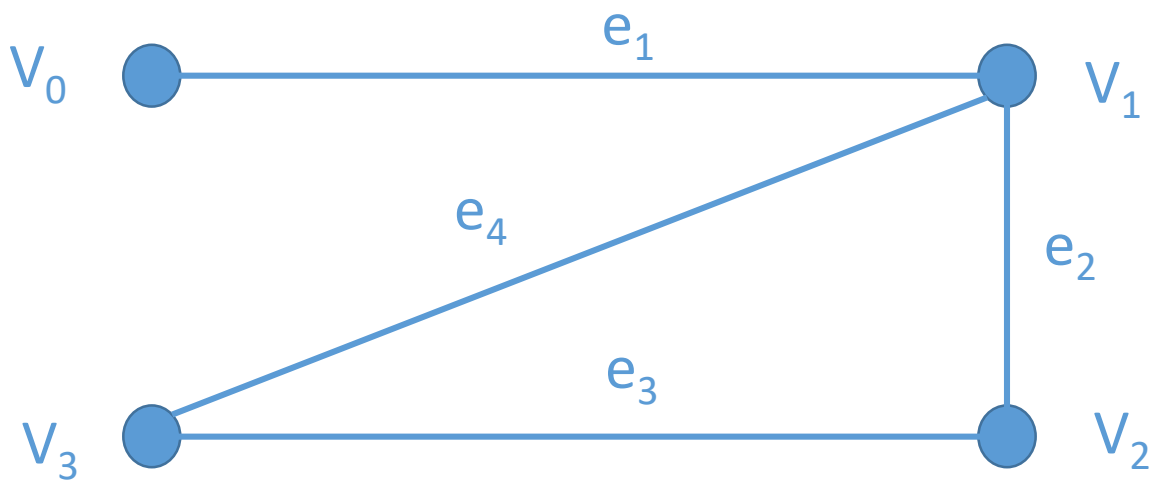
Corollary 8.4:

Any finite connected graph with two odd vertices is traversable.

Euler circuit and Euler Path

Definition

An Euler circuit in a graph G is a simple circuit containing every vertex has even degree of G. An Euler path in G is a simple path containing every edge of G.

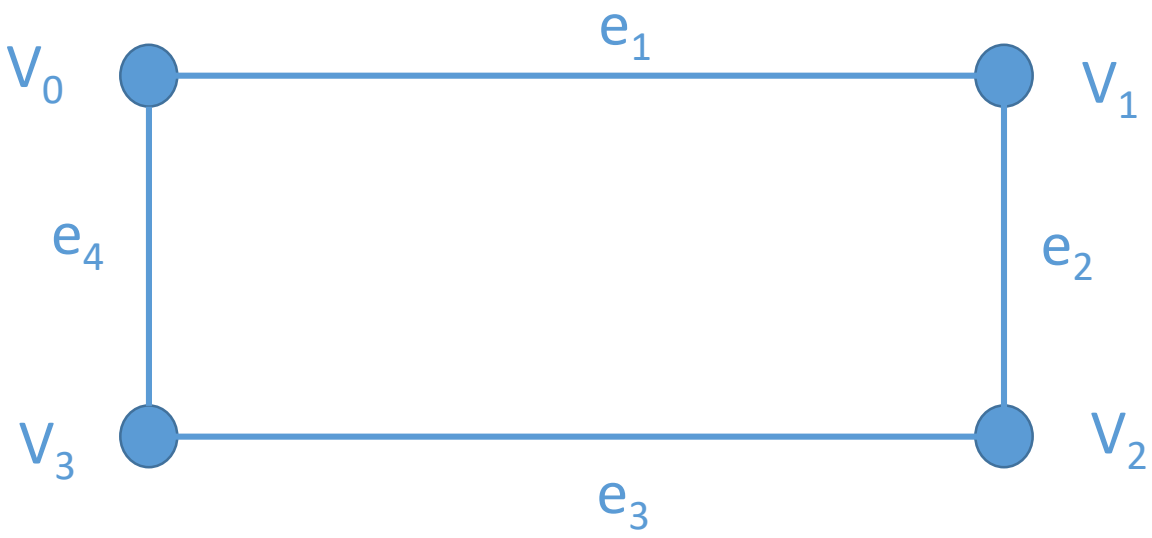


Euler circuit and Euler Path

Definition

An Euler circuit in a graph G is a simple circuit containing every edge of G.

An Euler path in G is a simple path containing every edge of G.



Problem - 1

Which of the undirected graphs in Figure 3 have an Euler circuit? Of those that do not, which have an Euler path?

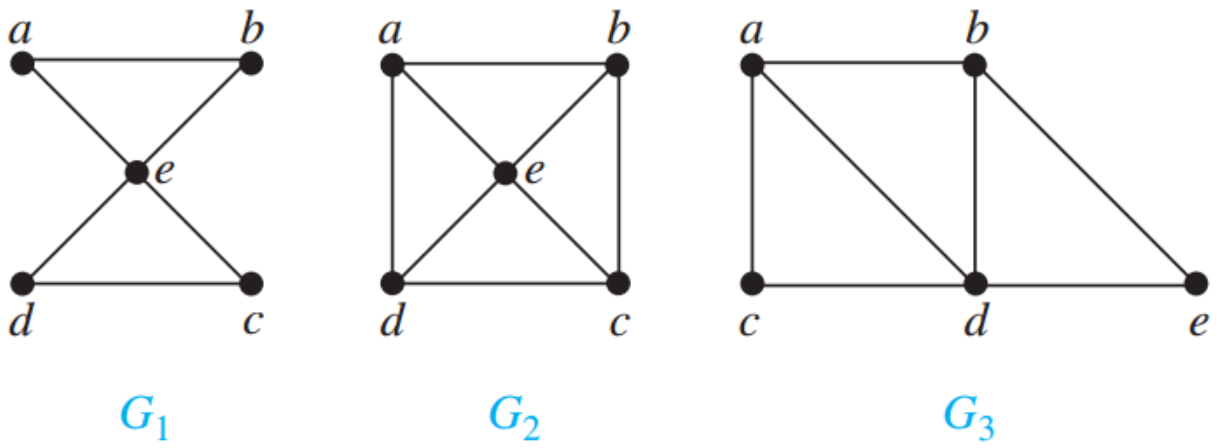


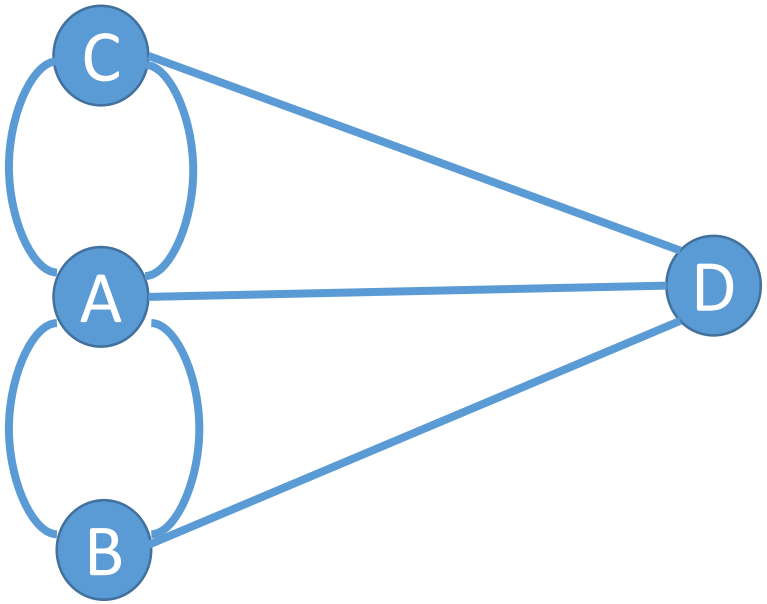
FIGURE 3 **The Undirected Graphs G_1, G_2, and G_3.**

Graph	Euler Path	E. Circuit
G_1	1	1
G_2	0	0
G_3	1	0

Euler path and circuit

Theorem 1

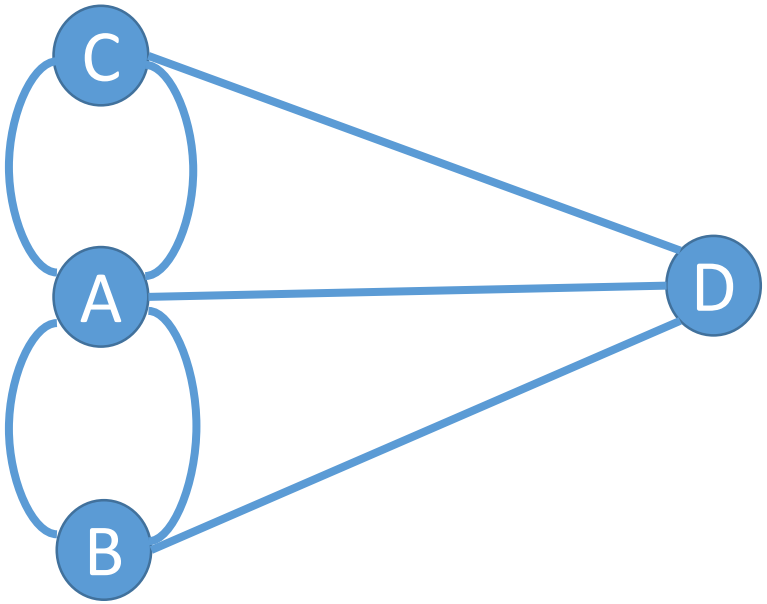
A connected multigraph with at least two vertices has an ***Euler circuit*** if and only if each of its vertices has even degree.



Euler path and circuit

Theorem 2

A connected multigraph has an ***Euler path*** but ***not an Euler circuit*** if and only if it has exactly two vertices of odd degree.



Problem - 3

Which graphs shown in Figure have an Euler path?

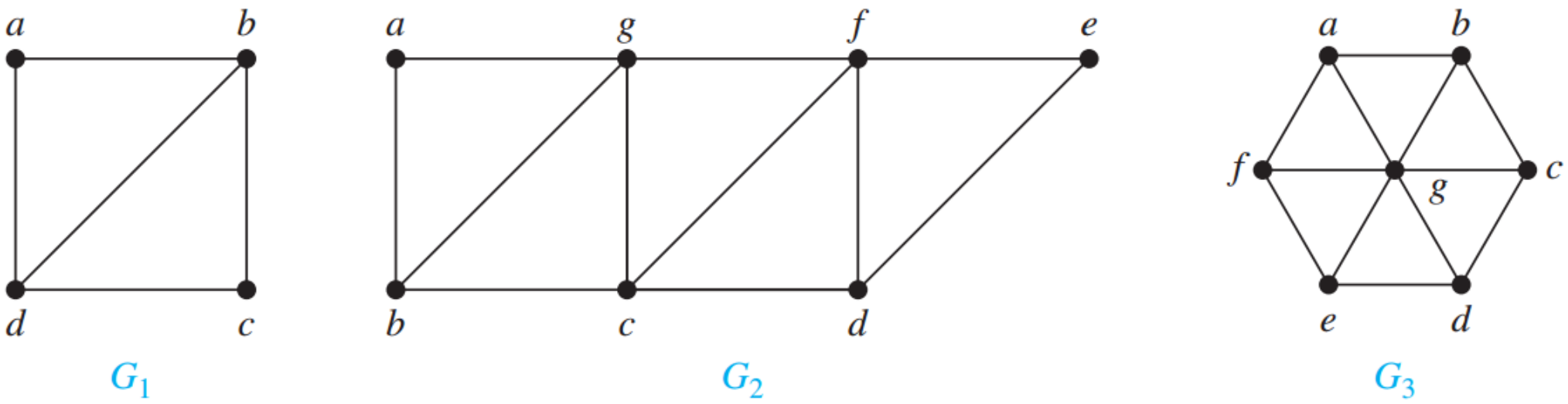


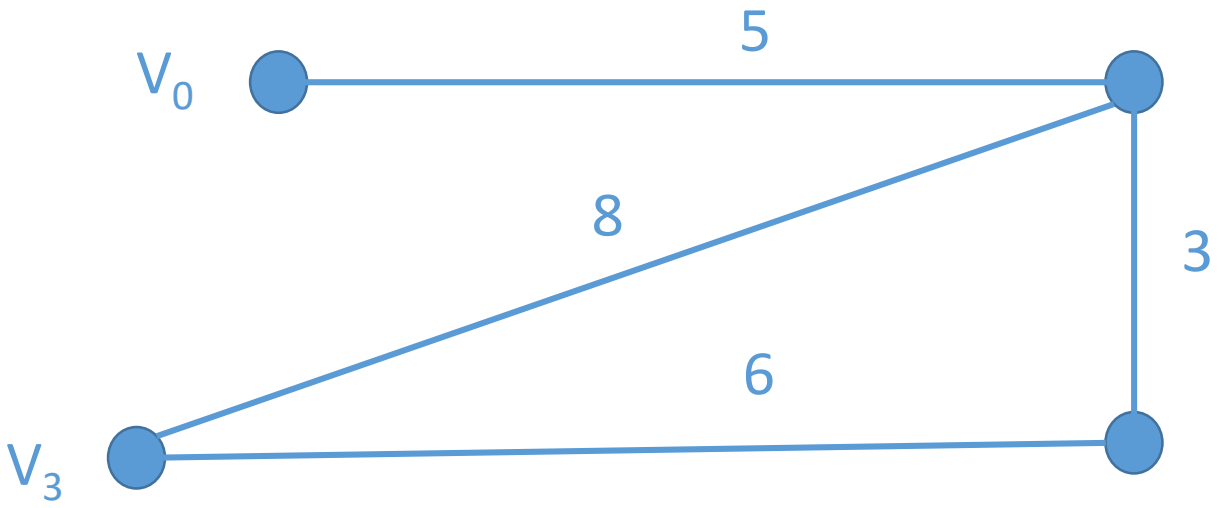
FIGURE 7 **Three Undirected Graphs.**

Shortest Path Problem

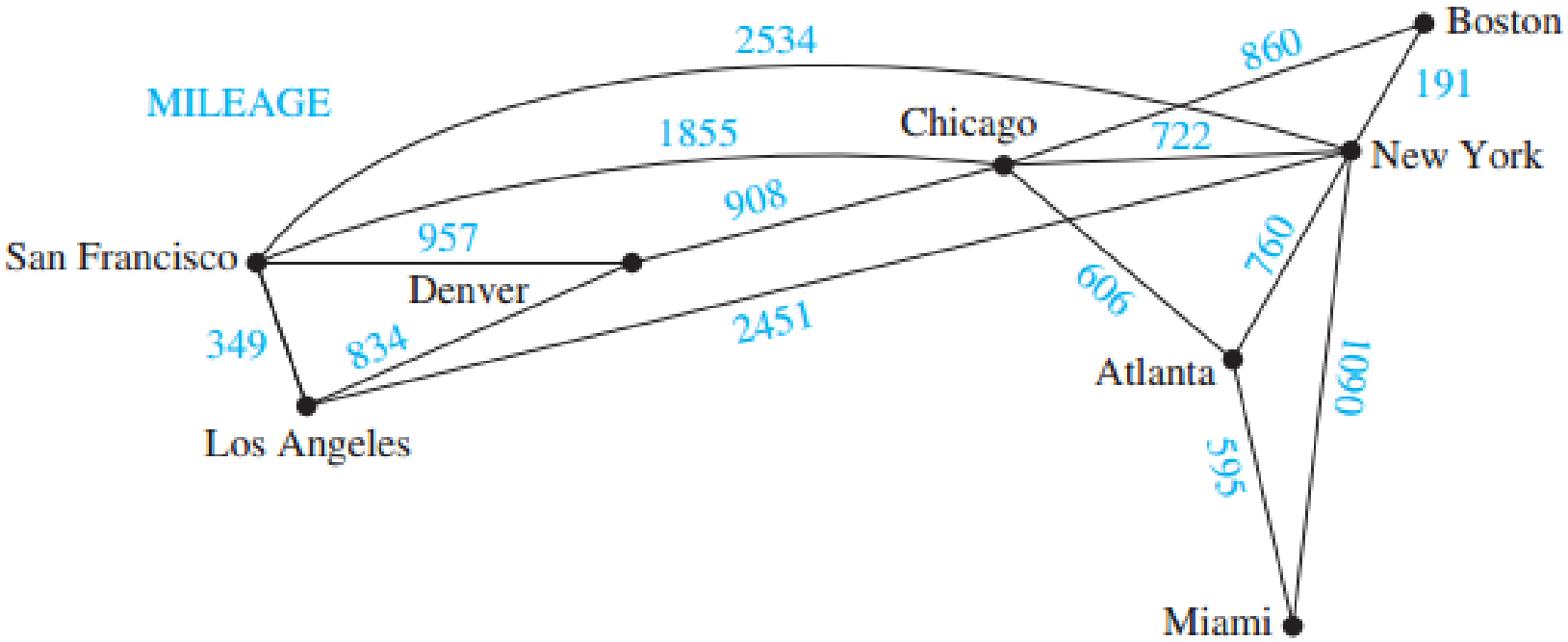
Weighted Graph

Definition

Graphs that have a number assigned to each edge are called weighted graphs.



Weighted Graph

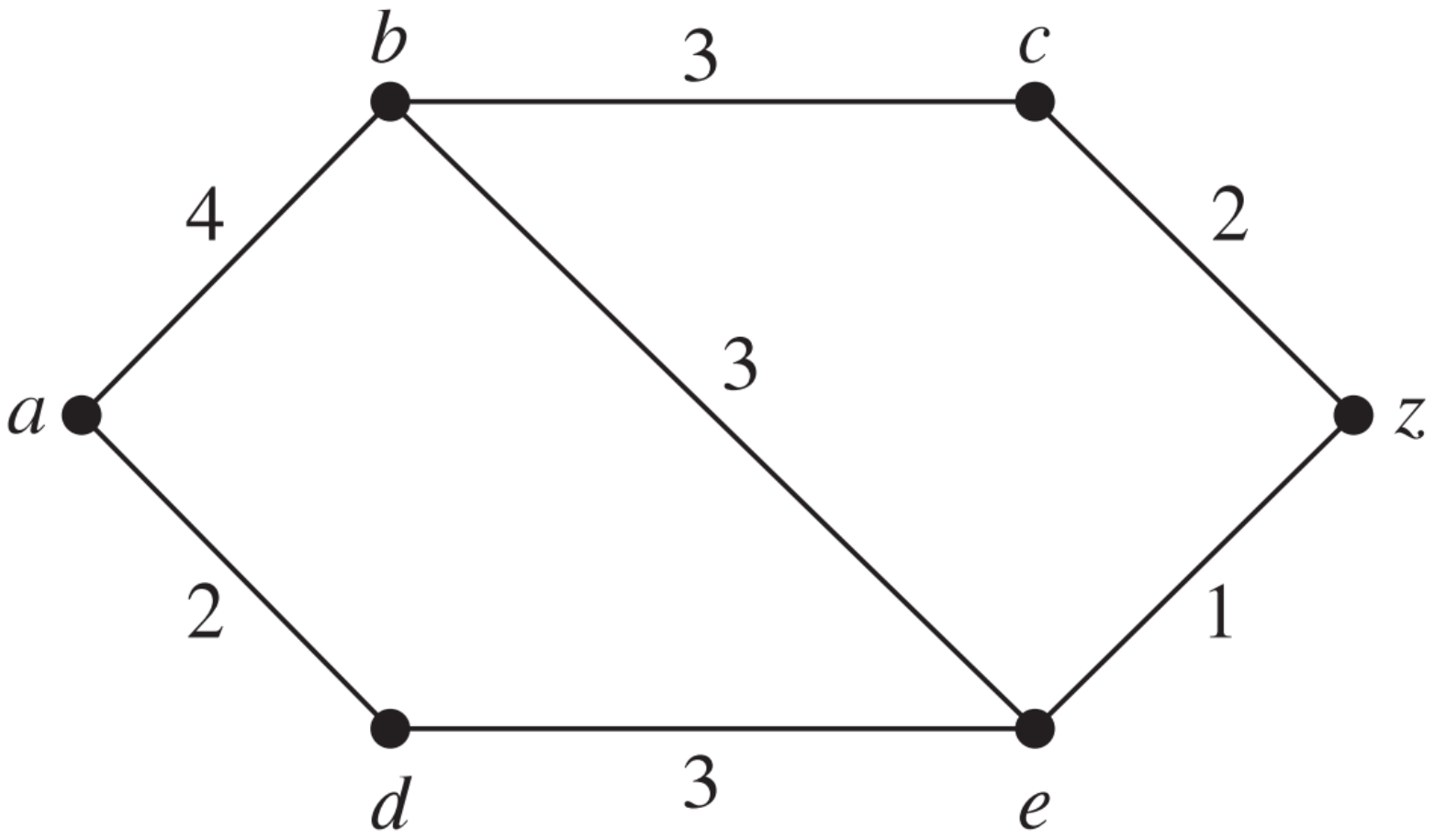


Shortest path algorithm

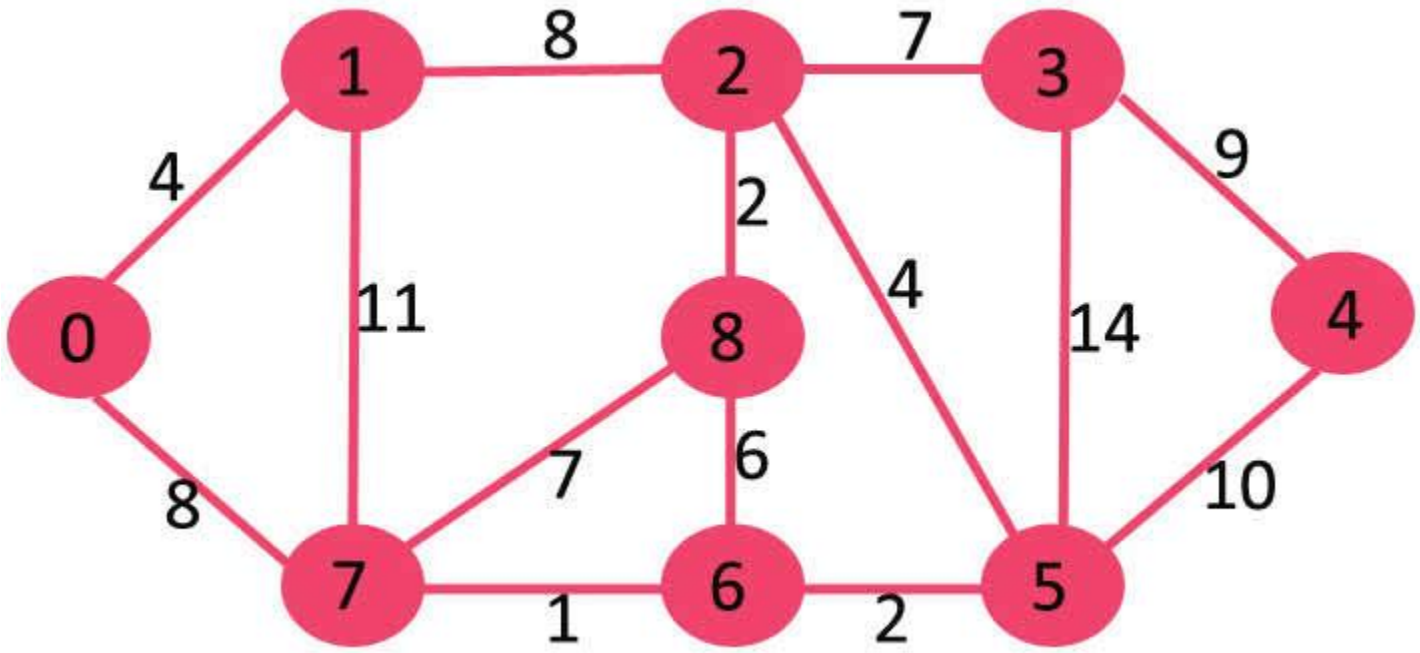
- Dijkstra Algorithm
- Bellman–Ford algorithm
- A* search algorithm
- Floyd–warshall Algorithm

Dijkstra's algorithm is an algorithm for finding the shortest paths between nodes in a graph, which may represent, for example, road networks. It was invented by computer scientist Edsger W. Dijkstra in 1956 and published three years later.

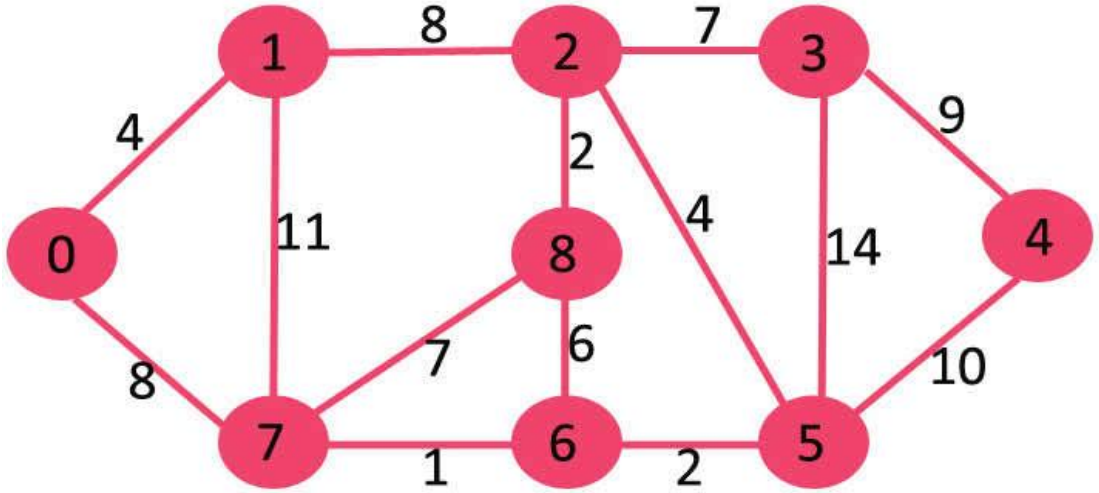
Dijkstra's algorithm

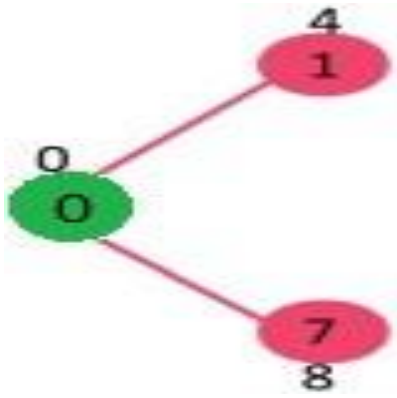


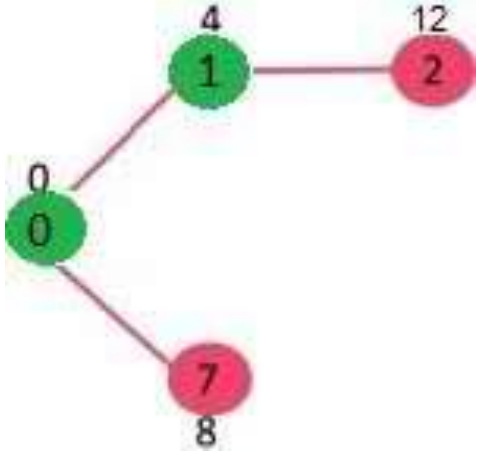
Dijkstra's algorithm

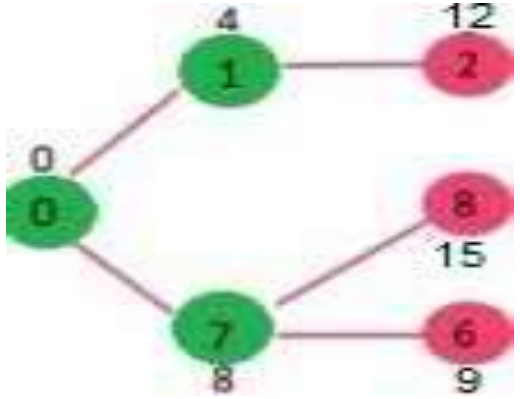


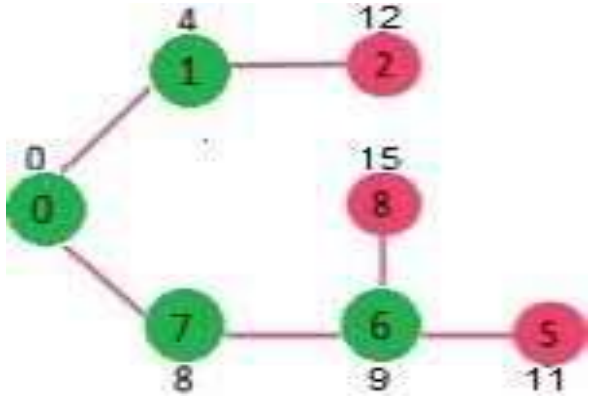
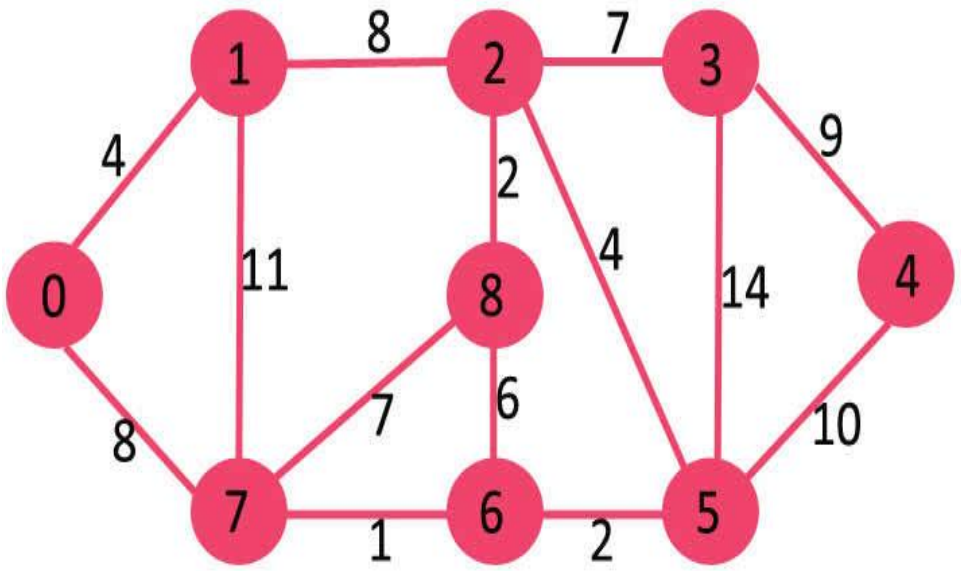
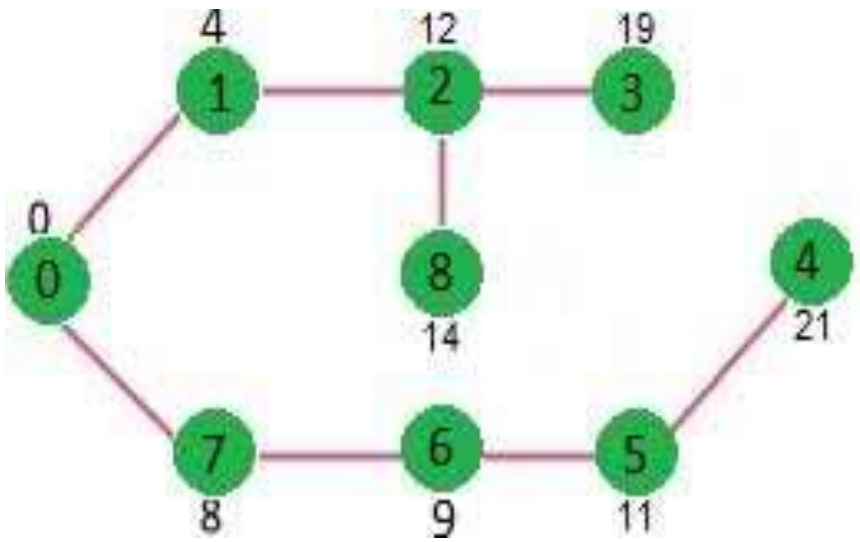
Dijkstra's algorithm











Dijkstra's algorithm